Supplementary material

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S1. Paired household-level tests

Table S1 lists every paired comparison. Each household's test MAE was averaged over the seeds of an arm before testing, so each test has one pair per household (100 pairs). The Holm correction was applied over all rows of the table together. Median ΔMAE is the median over households of the difference A minus B; negative values favour A.

Table S1. Wilcoxon signed-rank tests over households (seed-averaged MAE). A better: share of households with lower MAE under A.

A	B	median ΔMAE (A−B)	A better (%)	p	p_{Holm}
FedAvg, α=0.3	Local-only	+0.0012	34	0.0005	0.0047
FedAvg, α=0.3	Centralised	+0.0025	0	$<10^{-4}$	$<10^{-4}$
FedAvg, random	Centralised	+0.0021	0	$<10^{-4}$	$<10^{-4}$
FedProx, α=0.3	FedAvg, α=0.3	−0.0001	69	0.0007	0.0061
FedProx, cluster	FedAvg, cluster	+0.0001	31	0.0004	0.0047
FedProx, α=0.3, partial	FedAvg, α=0.3, partial	+0.0003	19	$<10^{-4}$	$<10^{-4}$
HA-FedProx, α=0.3	FedAvg, α=0.3	+0.0001	42	0.0674	0.2021
HA-FedProx, α=0.3	FedProx, α=0.3	+0.0001	43	0.0031	0.0217
HA-FedProx, cluster	FedAvg, cluster	+0.0003	25	$<10^{-4}$	$<10^{-4}$
HA-FedProx, cluster	FedProx, cluster	+0.0001	39	0.0012	0.0093
HA-FedProx, α=0.3, partial	FedAvg, α=0.3, partial	+0.0004	22	$<10^{-4}$	$<10^{-4}$
HA-FedProx, α=0.3, partial	FedProx, α=0.3, partial	+0.0002	35	0.0469	0.1875
FedAvg, cluster	FedAvg, random	+0.0001	44	0.0106	0.0638
FedAvg, K=100	Local-only	+0.0027	18	$<10^{-4}$	$<10^{-4}$
FedAvg, K=100	Centralised	+0.0045	0	$<10^{-4}$	$<10^{-4}$
HA-FedProx, K=100	FedAvg, K=100	+0.0003	22	$<10^{-4}$	$<10^{-4}$
FedAvg, α=0.3, fp16	FedAvg, α=0.3	−0.0001	92	$<10^{-4}$	$<10^{-4}$
Centralised	Local-only	−0.0016	72	$<10^{-4}$	$<10^{-4}$
FedAvg, K=1, random	Centralised	−0.0000	52	0.1046	0.2092
FedAvg, random	FedAvg, K=1, random	+0.0023	1	$<10^{-4}$	$<10^{-4}$
FedAvg, α=0.3 + local FT	Local-only	−0.0001	51	0.4960	0.4960
FedAvg, α=0.3 + local FT	Centralised	+0.0013	26	$<10^{-4}$	$<10^{-4}$
FedAvg, K=100 + local FT	Local-only	+0.0006	39	0.0142	0.0711

S2. Training budget and round selection

Table S2 gives, for each number of clients, the local Adam steps per client and round, the round chosen on validation error and the test MAE, for the 30-round budget of the main text and for runs of up to 100 rounds that stopped after 25 rounds without a new best validation MAE. Fig. S1 shows the validation curves.

Table S2. Budget per number of clients K (FedAvg, random grouping). Steps/round: Adam steps per client per round. Best rd.: round selected on validation MAE (mean over seeds). Gap: test MAE relative to centralised.

K	Steps/round	Best rd. (30)	MAE (30)	Gap (%)	Best / run (100)	MAE (100)	Gap (%)
1	4,534	16.3	0.0808	−0.3	–	–	–
5	907	23.0	0.0827	+2.0	–	–	–
20	227	22.0	0.0854	+5.3	96 / 100	0.0832	+2.7
50	91	26.7	0.0869	+7.2	56 / 76	0.0858	+5.9
100	46	29.0	0.0894	+10.3	95 / 100	0.0871	+7.4

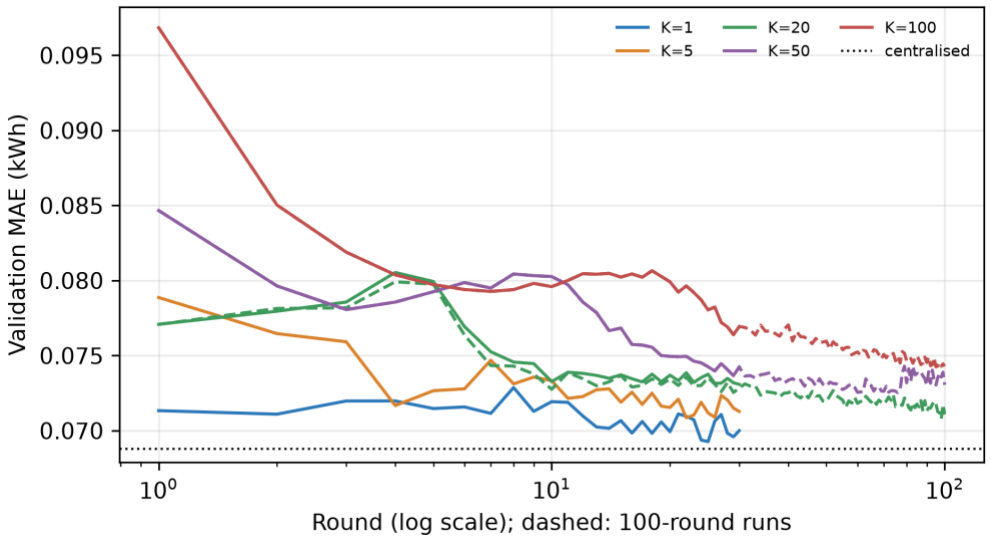


Figure S1. Validation MAE per round for each number of clients (mean over seeds). Solid: 30-round runs; dashed: runs of up to 100 rounds. Dotted: centralised.

S3. Heterogeneity-adaptive proximal coefficient

The original submission proposed a rule that sets the proximal coefficient from the disagreement between client updates. After round r the server holds the updates $\Delta_k = w_k - w_r$ and their weighted mean $\overline{\Delta}$, and computes

$$\rho_r = \frac{\sum_k p_k \|\Delta_k - \overline{\Delta}\|^2}{\sum_k p_k \|\Delta_k\|^2} \in [0,1], \mu_{r+1} = \text{clip}(\mu_0 \rho_r,\ 10^{-4},\ 1) \quad \text{(S1)}$$

The rule never improved on FedAvg. Under the cluster partition it was worse for 75 % of households ($p_{\text{Holm}} < 10^{-4}$). Its signal ρ_r ranked partitions by heterogeneity (Spearman +0.51 with H_{JS}, $p = 0.0044$, $n = 30$ runs), but on average it rose from 0.15 in round 1 to 0.96 in the last round, a change about 46 times its spread across partitions. As the shared descent direction shrinks during training, any residual disagreement pushes ρ_r towards one, so a coefficient proportional to it grows late in training whatever the data. Table S3 gives all three aggregation rules for each grouping.

Table S3. All aggregation rules for each grouping at $K = 20$ (three seeds). H: composite index of the original submission; $\bar{\rho}$: mean update disagreement.

Partition	Method	H	H_{JS}	$\bar{\rho}$	Test MAE	Gap to central (%)
α=0.1	FedAvg	0.241	0.0189	0.874	0.0856 ± 0.0004	+5.6
α=0.1	FedProx	0.242	0.0181	0.863	0.0856 ± 0.0006	+5.6
α=0.1	HA-FedProx	0.242	0.0181	0.844	0.0859 ± 0.0003	+5.9
α=0.3	FedProx	0.243	0.0153	0.861	0.0858 ± 0.0001	+5.9
α=0.3	HA-FedProx	0.243	0.0153	0.844	0.0863 ± 0.0003	+6.5
α=0.3	FedAvg	0.246	0.0152	0.871	0.0860 ± 0.0003	+6.0
random	FedAvg	0.253	0.0106	0.858	0.0854 ± 0.0008	+5.3
random	FedProx	0.255	0.0105	0.845	0.0855 ± 0.0003	+5.4
random	HA-FedProx	0.255	0.0105	0.822	0.0859 ± 0.0009	+6.0
α=1.0	FedAvg	0.262	0.0125	0.868	0.0851 ± 0.0007	+5.0
α=1.0	FedProx	0.270	0.0124	0.856	0.0857 ± 0.0001	+5.7
α=1.0	HA-FedProx	0.270	0.0124	0.841	0.0857 ± 0.0008	+5.8
cluster	FedProx	0.381	0.0202	0.866	0.0858 ± 0.0002	+5.8
cluster	HA-FedProx	0.381	0.0202	0.851	0.0865 ± 0.0002	+6.7
cluster	FedAvg	0.381	0.0202	0.876	0.0857 ± 0.0003	+5.7

S4. Proximal coefficient grid

Table S4 gives the best validation MAE for each coefficient (cluster partition, seed 42). The row $\mu = 0$ is plain FedAvg.

Table S4. Best-round validation MAE (kWh) over the proximal coefficient grid.

μ (FedProx) or μ_0 (adaptive)	FedProx	Adaptive (HA-FedProx)
0	0.07238	0.07238
0.0001	0.07260	0.07260
0.001	0.07259	0.07268
0.01	0.07320	0.07332
0.1	0.07666	0.07682
1	–	0.07803

S5. Additional cohorts

Two further cohorts of 100 homes were drawn with the same eligibility rules and stratification but different selection seeds (43 and 44); they share three or four homes with the main cohort. Each arm ran with two seeds.

Table S5. Test MAE (kWh) of the main arms on three cohorts, and the FedAvg gap to centralised training at $K = 1$, 5, 20 and 100.

Cohort	Centralised	Local-only	FedAvg, K=1	FedAvg, K=5	FedAvg, K=20	FedAvg, K=100	FedAvg gap, K=1/5/20/100 (%)
main (seed 42)	0.0812	0.0838	0.0806	0.0828	0.0855	0.0898	−0.7, +2.0, +5.4, +10.7
cohort 43	0.0847	0.0878	0.0832	0.0853	0.0895	0.0909	−1.7, +0.8, +5.7, +7.4
cohort 44	0.0876	0.0866	0.0856	0.0866	0.0884	0.0910	−2.3, −1.2, +0.9, +3.8

S6. Error by consumption quintile

Households were divided into quintiles of mean training load (Q1 lowest). Table S6 gives the median household test MAE in each quintile, Fig. S2 the distribution of the benefit of each arm over local training, and Table S7 summary statistics of that benefit.

Table S6. Median household test MAE (kWh) by consumption quintile. FedAvg at $K = 20$ unless stated.

Quintile	Local-only	Centralised	FedAvg, α=0.3	FedAvg, α=0.3 + FT	FedAvg, K=100
Q1	0.0279	0.0281	0.0299	0.0297	0.0304
Q2	0.0467	0.0470	0.0494	0.0481	0.0514
Q3	0.0755	0.0742	0.0760	0.0782	0.0790
Q4	0.1053	0.1031	0.1099	0.1073	0.1117
Q5	0.1393	0.1317	0.1408	0.1393	0.1487

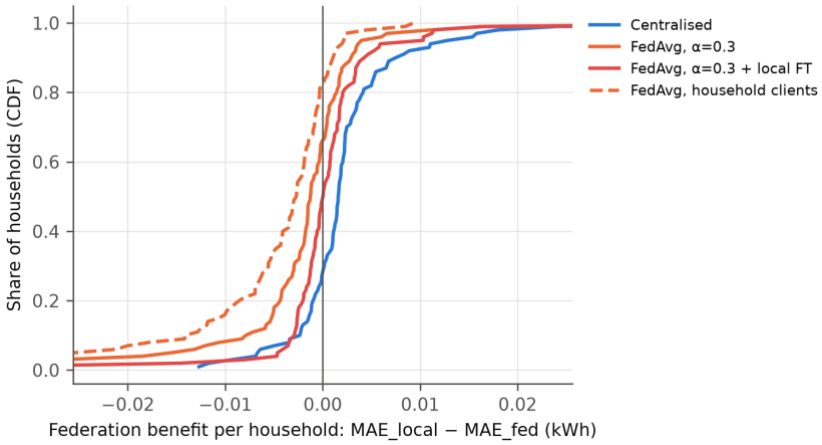


Figure S2. Distribution over households of the benefit of each collaborative arm over local training ($MAE_{local} - MAE_{arm}$, kWh; left of zero means worse off).

Table S7. Equity relative to local-only training (seed-averaged household MAE): share harmed (any difference), median benefit, 90th-percentile household MAE and share harmed after fine-tuning (FT).

Arm	Harmed vs local (%)	Median benefit (kWh)	P90 MAE	Harmed after FT (%)
Centralised	28	+0.0016	0.1364	–
FedAvg, random	56	−0.0007	0.1436	51
FedAvg, α=0.3	66	−0.0012	0.1442	49
FedAvg, cluster	60	−0.0008	0.1431	46
HA-FedProx, α=0.3	66	−0.0013	0.1454	55
FedAvg, K=100	82	−0.0027	0.1514	61
HA-FedProx, K=100	85	−0.0028	0.1515	67

S7. Calendar inputs and scaled error

Sine and cosine encodings of time of day and day of week were added as input channels to every arm in Table S8, with all other settings unchanged.

Table S8. Effect of calendar inputs, and MASE (scaled per household by the in-sample one-step naive MAE). Gain: MAE reduction relative to persistence; cal.: with calendar inputs. Gap: MAE relative to centralised training with the same inputs.

Arm	MAE	MASE	Gain (%)	MAE, cal.	MASE, cal.	Gain, cal. (%)	Gap, cal. (%)
Centralised	0.0811	0.987	+8.9	0.0815	0.996	+8.4	–
Local-only	0.0840	1.006	+5.6	0.0838	1.007	+5.8	–
FedAvg, K=20	0.0854	1.045	+4.0	0.0863	1.057	+3.1	+5.8
FedAvg, K=100	0.0894	1.111	−0.5	0.0919	1.158	−3.3	+12.8